Elastic Observability

How Elastic is redefining observability for the AI era: from datastore to autonomous SRE.

elastic

The Search AI Company



Meet Search AI

The fusion of **search** and **artificial intelligence**

OBSERVABILITY · THE AI ERA

The challenge you know.
At a scale you've never seen.

WHAT WE BUILT FOR → Microservices complexity | Multi-cloud sprawl | High change velocity | Manual toil | Alert fatigue | Observability cost

THEN AI ARRIVED. AND MULTIPLIED EVERY PROBLEM.

01 · DEVELOPMENT

×100

Agents generate code in parallel, around the clock

- Before: one team, one pipeline, human review
- Now: **swarms of agents** committing every second
- Ships without review.
  Code no one has read.
- Unknown dependencies, unknown risk surface
- More pipelines =
  more instrumentation cost

02 · TESTING & STAGING

×10

Infra multiplies as fast as agents can provision it

- Before: a handful of pods, one staging env
- Now: **agents spin up dozens of environments** in parallel
- Staging and prod are
  permanently out of sync
- Infrastructure grows faster than teams can track
- Ephemeral envs multiply
  observability spend

03 · PRODUCTION

?×

Unread code ships fast. Incidents follow.

- **Incident rate accelerates** with deployment frequency
- SREs paged for
  code they've never seen
- Alert volume explodes. Runbooks are obsolete.
- Debugging with
  zero tribal knowledge
- 10× services =
  10× logs, metrics, traces and the bill

THE RESULT

More code. More infra. More signals.

Faster failures. Zero context. No runbook.

Observability bill grows as fast as the problem.

Observability built for humans can't keep up.

ELASTIC OBSERVABILITY

Three pillars. **One platform.**

01 · PILLAR

Streams

The telemetry pipeline for all your data

Open-standard ingestion at any scale. Whether you're running OTel, Prometheus, or any custom source — Streams is the universal pipeline that gets your data in, routed, and stored efficiently.

OpenTelemetry · Prometheus · Any Data · Open Standards

02 · PILLAR

Signals

Every observability experience, unified

All the ways your team makes sense of systems — Logs, Metrics, Traces, Synthetics, and Profiles — in one place. Purpose-built experiences for every signal type, on a single platform.

Logs · Metrics · Traces · Synthetics · Profiles

03 · PILLAR

Nightshift

Fix it before you wake up

The autonomous AI SRE. Nightshift investigates incidents, runs playbooks, and remediates — around the clock, without waiting for a human to be paged.

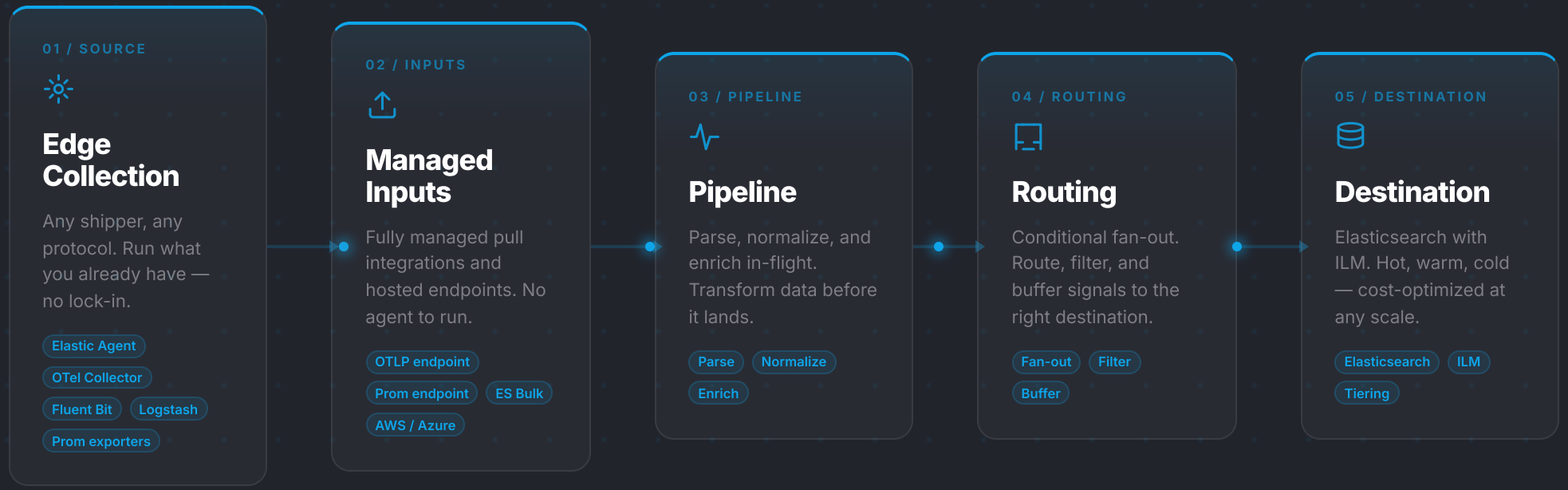
AI SRE · Auto-Remediation · RCA · Agentic

Together, these three pillars define the Observability roadmap — from open-standard data pipeline, to unified signal experiences, to autonomous AI action.

01 · PILLAR · STREAMS

Streams

Five stages. One pipeline. Every signal from any source to Elasticsearch.



ELASTIC OBSERVABILITY · OPENTELEMETRY

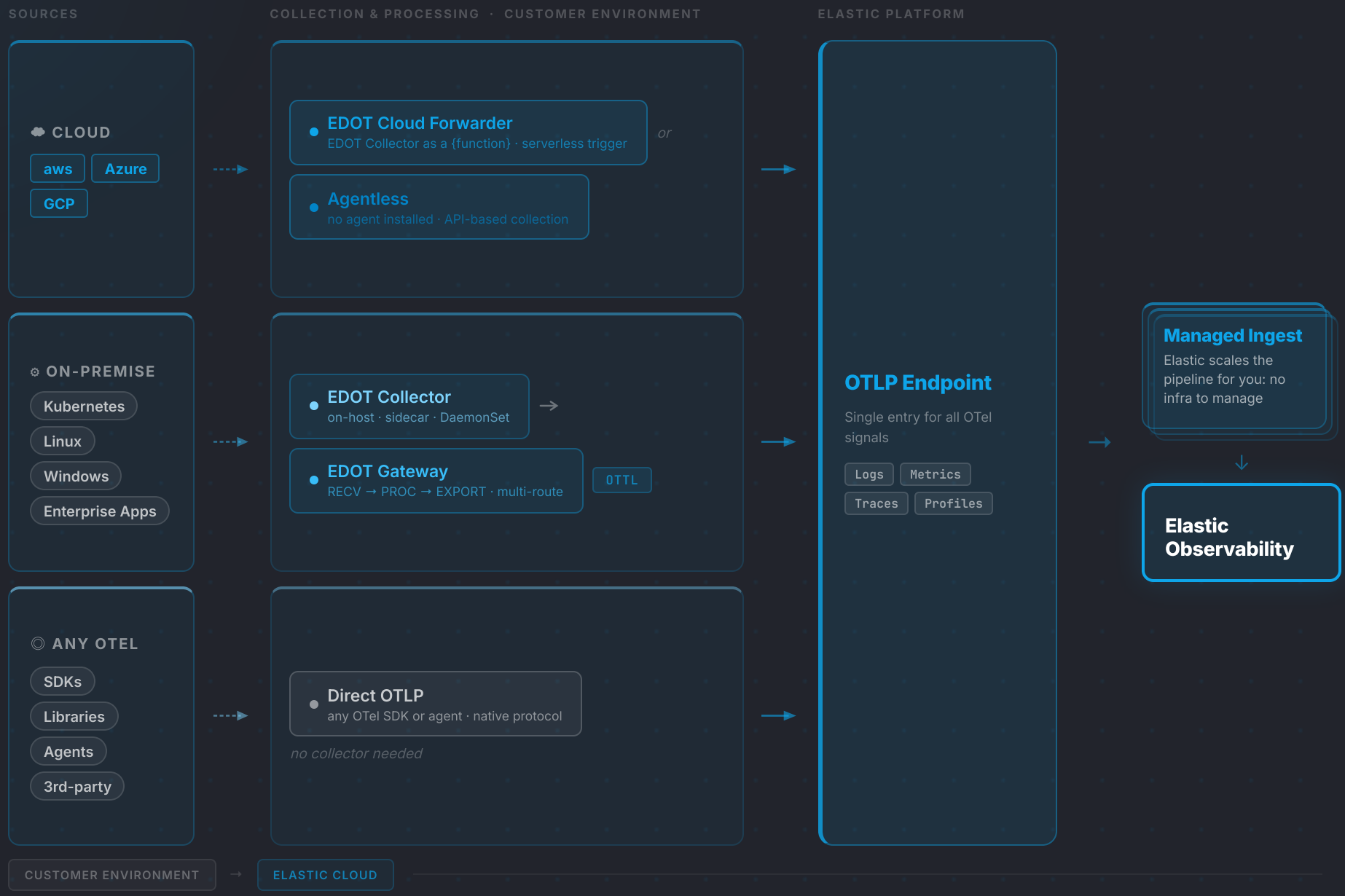
Collect everything, from everywhere

#1 OTel Contributor | Managed Ingest

Cloud, on-prem, or mixed: **EDOT meets your data wherever it lives** and gives you the flexibility to collect, process, or forward it at any scale.

- **Cloud Forwarder & Agentless**
  EDOT as a {function}: no infra, triggers on cloud events
- **EDOT Collector + Gateway**
  Edge collection with OTTL pipeline processing on-prem
- **Managed Ingest**
  Send direct: Elastic scales the ingest layer for you

#1 **Top OTel contributor**
We build the standard we support



ELASTIC OBSERVABILITY · STREAMS

Streams: From raw data to immediate insights

Always up-to-date · real-time · zero configuration

Data Collection
Any source, any format

- **OpenTelemetry** — OTLP · gRPC · HTTP
- **Kinesis Firehose** — AWS · streaming
- **FluentD / Bit** — Logs · structured events
- **Prometheus** — Metrics · scrape · remote write

Top Level Streams
Central input destinations

- `/logs`
- `/metrics`
- `/traces`

Sub Streams
Schema · lifecycle · access

- `/logs/aws/eks`
  - `/logs/aws/eks/nginx`
  - `/logs/aws/eks/kafka`
- `/metrics/k8s/nodes`
  - `/metrics/k8s/pods`
- `/traces/services/api`
  - `/traces/services/db`

KIs & Entities
Auto-detected from telemetry

- AWS · GCP · Azure
- k8s · EKS · GKE · Docker
- nginx · envoy · traefik · istio
- postgres · redis · mongodb · mysql
- kafka · rabbitmq · nats · pulsar
- prometheus · grafana · consul

Topology & dependencies auto-mapped in real time

Significant Events
AI-powered insights

- Severe errors & crashes
- Version updates
- Change in pattern distribution

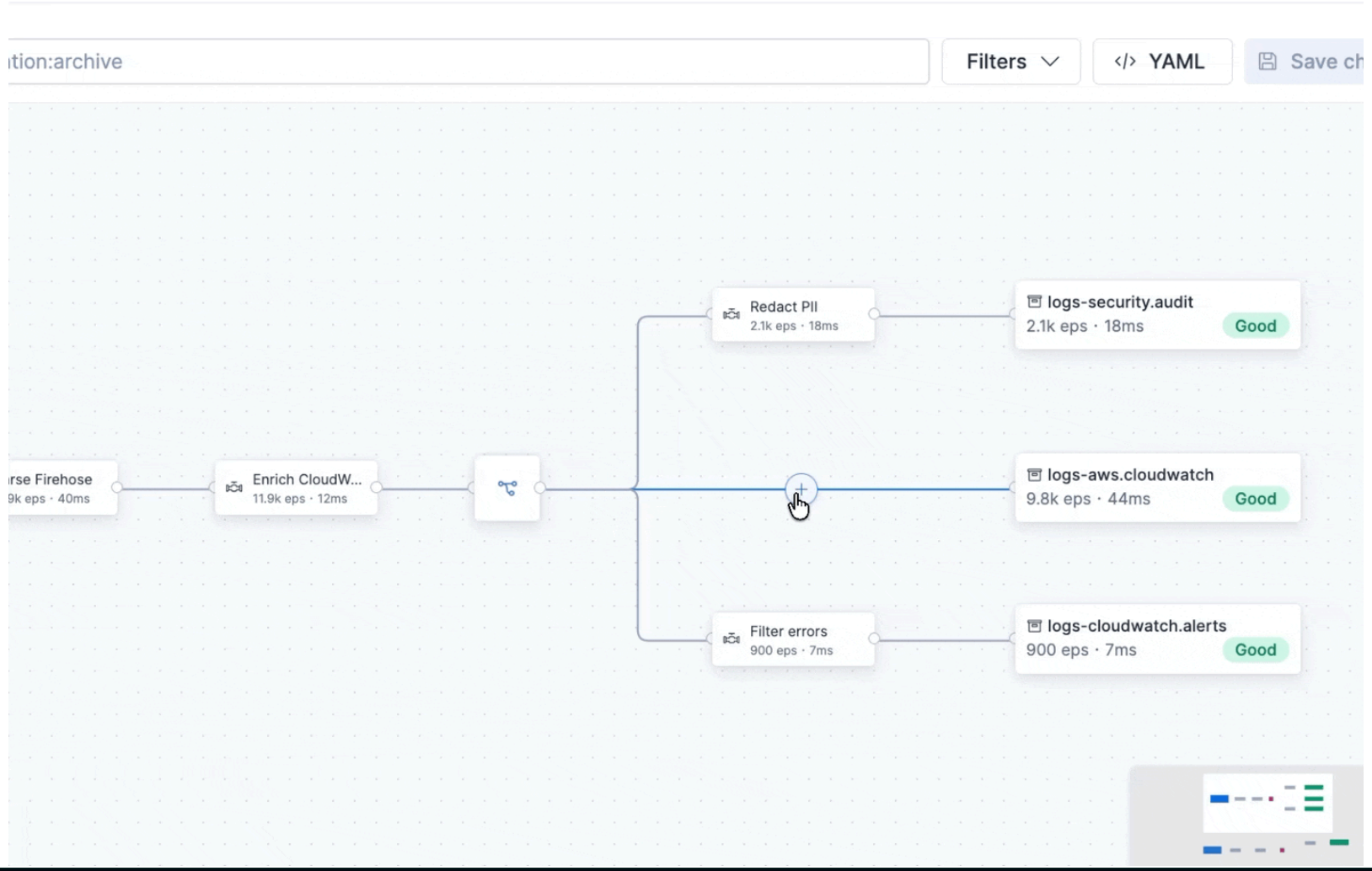
Generative AI extracts meaning from raw signals.

→ AGENT-READY

STREAMS · DEMO

Visual pipeline builder

Route, enrich, and filter data in real time — no YAML required.



02 · PILLAR · SIGNALS

Signals

Every observability experience on one platform — Logs, Metrics, Traces, Synthetics, and Profiles.

Logs

Structure, search, and analyze every log at scale.

- LogsDB — 65% smaller storage
- ES|QL for fast ad-hoc queries
- Pattern analysis & clustering

Metrics

Prometheus-native. Columnar storage. At any scale.

- Native PromQL support
- 30× faster than Prometheus
- Kubernetes & cloud out of the box

Traces

Distributed tracing and service dependency mapping.

- OTel-native ingestion
- Service map & topology
- Trace-to-log correlation

Synthetics

Proactive uptime and performance monitoring.

- Browser & API monitors
- SLA tracking & alerting
- Global monitoring locations

Profiles

Continuous profiling for CPU, memory, and latency.

- Always-on, low overhead
- Flamegraphs in context
- Correlated with traces & logs

DATASTORE EFFICIENCY

The most efficient store for logs and metrics

LogsDB and columnar metrics storage deliver dramatic cost reduction without sacrificing query speed.

65%	30×	~40%
log storage reduction	faster metrics vs Prometheus	faster query time

GTM FOCUS · METRICS

Winning the Prometheus migration

Native PromQL, remote write, and Kubernetes integrations make Elastic the obvious destination for teams moving off Prometheus.

PromQL	OTel	k8s
fully native	standard ingestion	out of the box

ELASTIC OBSERVABILITY · LOGS

Best in class efficiency for Logs

THREE COMPOUNDING ENGINEERING BETS

LogsDB Index Mode

• GA · 8.17+

A purpose-built index mode for log data. Smart sorting by `host.name` and `@timestamp` places similar records adjacent, dramatically improving compression. Synthetic `_source` reconstructs fields on demand: no redundant storage.

65% storage reduction vs standard Elasticsearch

50% TCO reduction for long-term log retention

30% additional savings from smart index sorting alone

Read the deep dive →

Columnar datastore mode

• IN DEVELOPMENT

By skipping inverted indexes and BKD trees entirely and using compressed binary doc-values, Elasticsearch achieves near-columnar storage density: optimized for high-volume analytics workloads where every byte counts.

168 B per record in doc-values mode

~5× improvement vs standard Elasticsearch (805 B/record)

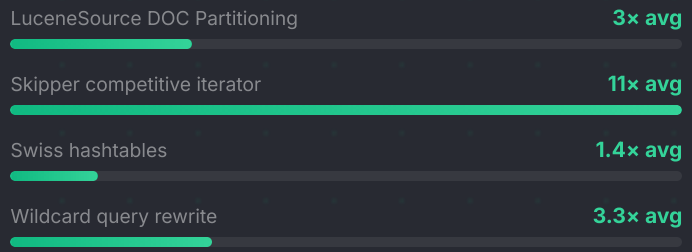
Near parity with best-in-class columnar stores: before full columnar mode

Full columnar logs will close the gap entirely

Query Latency: ~40% faster since Jan 2026

• ES 9.4

Three focused engineering investments have compounded across 9.x to dramatically cut average query execution time: and each improvement stacks on the last.



~40% faster average query time since Jan 2026

ENGINEERS DON'T CHANGE TOOLS — WE MEET THEM WHERE THEY ARE · 30× FASTER QUERIES VS PROMETHEUS · 4× LOWER TCO VS DATADOG · 0 DASHBOARD REWRITE

FY27 · BIGGEST LAUNCH IN YEARS

Metrics.

Metrics, logs, and traces — **one platform, one bill.**
30× faster. 4× lower cost. PromQL-native when you need the bridge.

June 29, 2026



ELASTIC OBSERVABILITY · METRICS

Best in class efficiency for Metrics

ELASTICSEARCH 9.4 BENCHMARKS

Blazing Fast Query Performance

30× faster: Prometheus & Mimir

- up to 30× faster — vs Prometheus
- up to 30× faster — vs Mimir
- 7× faster — vs ClickHouse

WHY IT MATTERS

Delivers faster AI investigations.

GA — Blazing fast: 30× vs Prom, Mimir

Significant Storage Efficiency Gain

6.6–8× overall vs ES 8.x

- 2.6× better — vs Prometheus
- 2× better — vs ClickHouse

WHY IT MATTERS

Faster root cause. Store more for richer AI context at lower cost.

GA IN 9.4

Prometheus-native Ingest & PromQL

on par

- on par — with Mimir (404K) & Prometheus
- 1.4× better — vs ClickHouse

WHY IT MATTERS

Zero friction to switch. Engineers using Grafana feel right at home.

TECH PREVIEW

Migrate in 1 day from Datadog and Grafana → Elastic

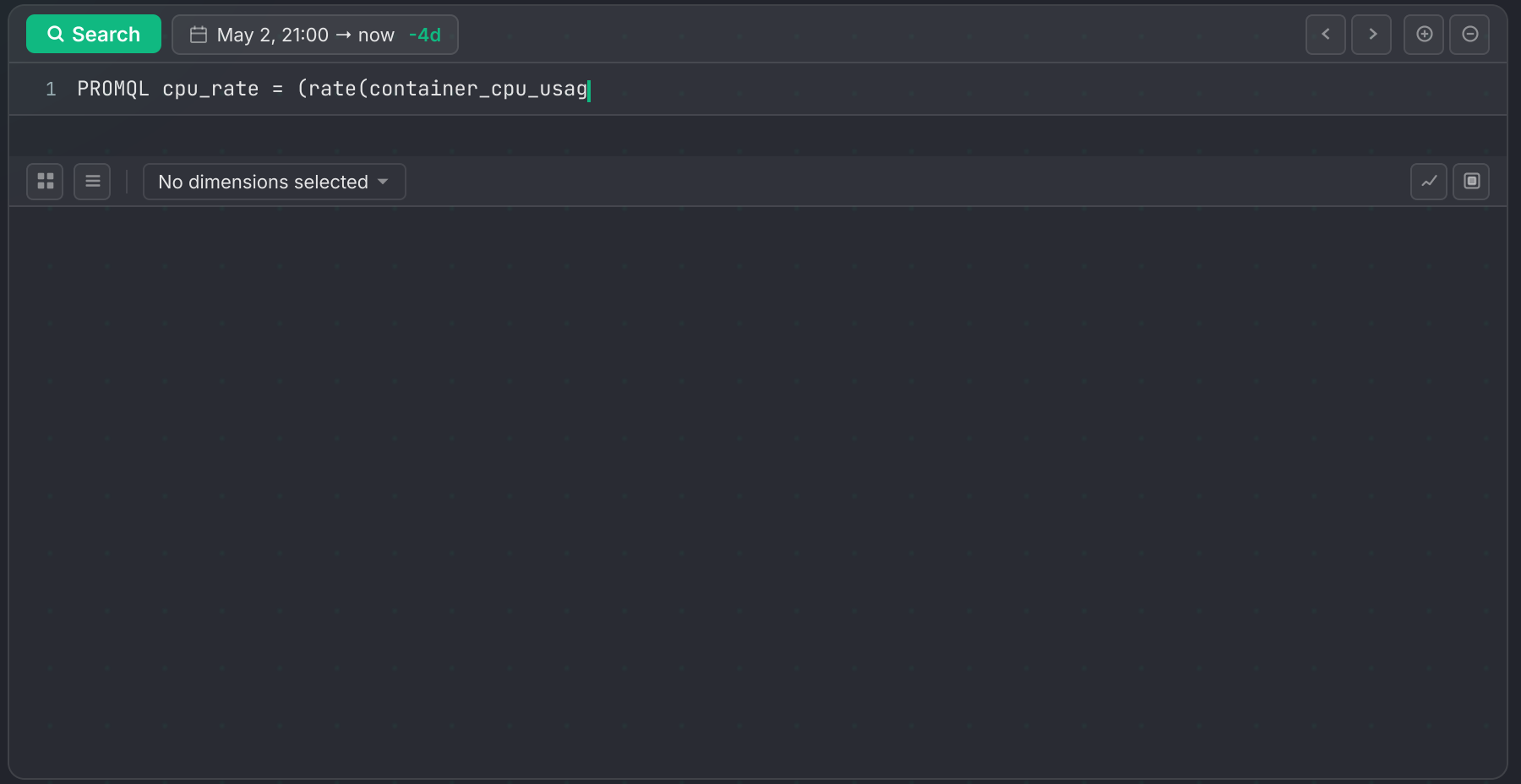
Your SREs already speak PromQL.

Now Kibana does too.

80%

of queries in the top 100 Grafana dashboards

natively executed in Kibana



Native in Kibana

No plugins, no adapters. Runs wherever ES|QL is supported.

Rich Analytics

rate, irate, sum by, max_over_time — works out-of-the-box.

up to 30× Faster

Same queries as Prometheus with significantly better performance.

elastic

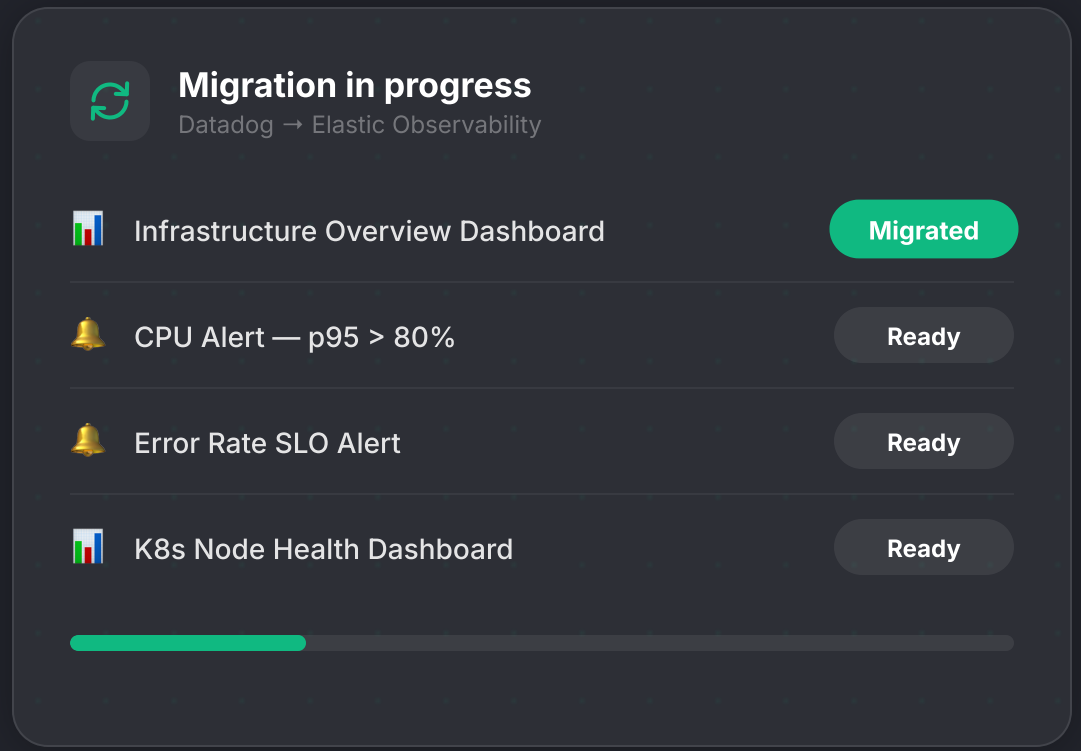
Migrate from Datadog or Grafana overnight.

Automatically converts your dashboards and alerting rules from Datadog and Grafana into Elastic — dramatically reducing cost and complexity.

1. **Connect your existing platform**
   Point the tool at your Datadog org or Grafana instance using API credentials.

2. **Automatic conversion**
   Dashboards, panels, alert rules, and PromQL queries are translated to Elastic equivalents.

3. **Go live in Kibana**
   Your team is fully operational in Elastic with familiar workflows intact — same day.



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A Lovable Kubernetes OOTB Experience

The most widely deployed cloud infrastructure

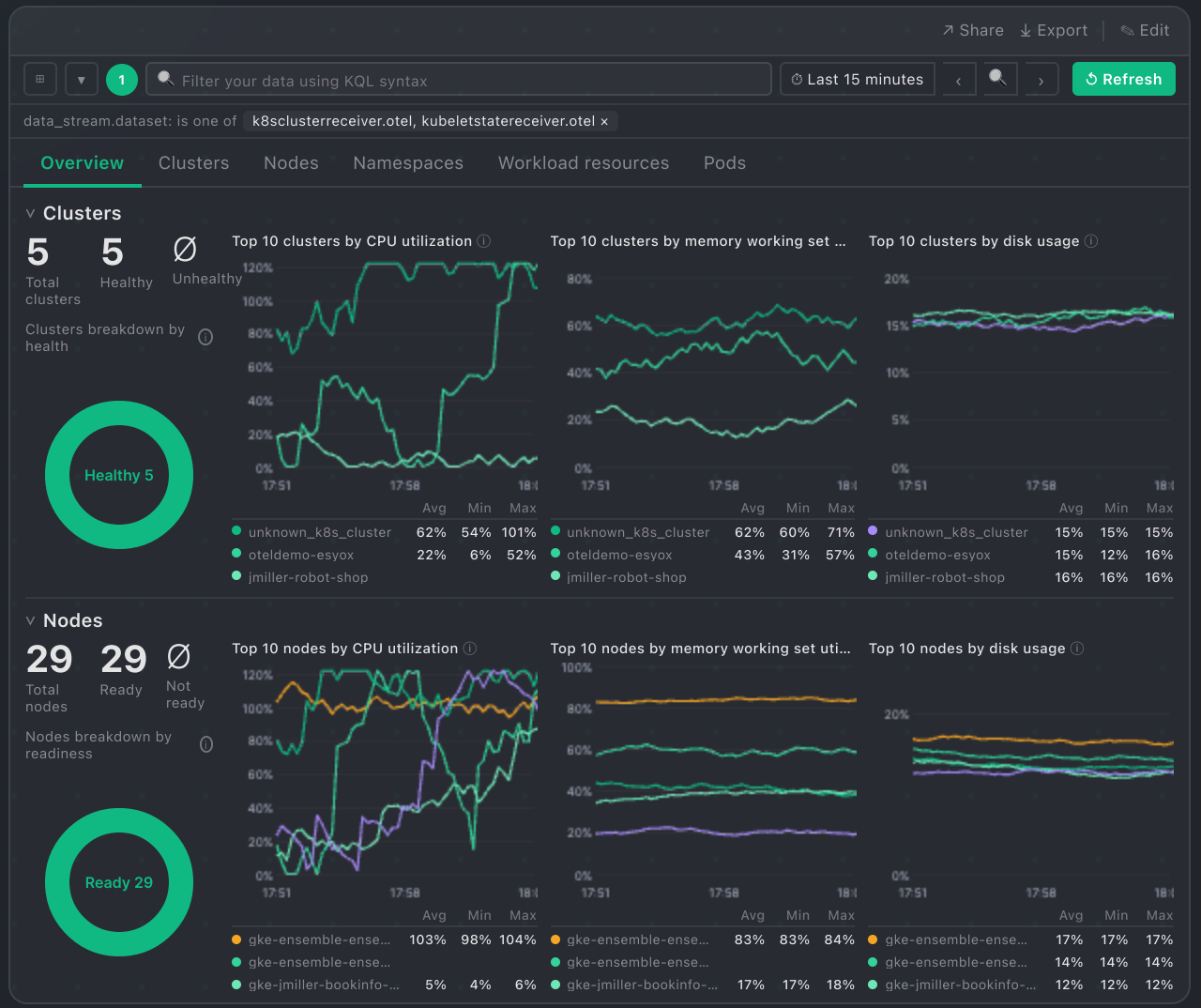
Why It Matters

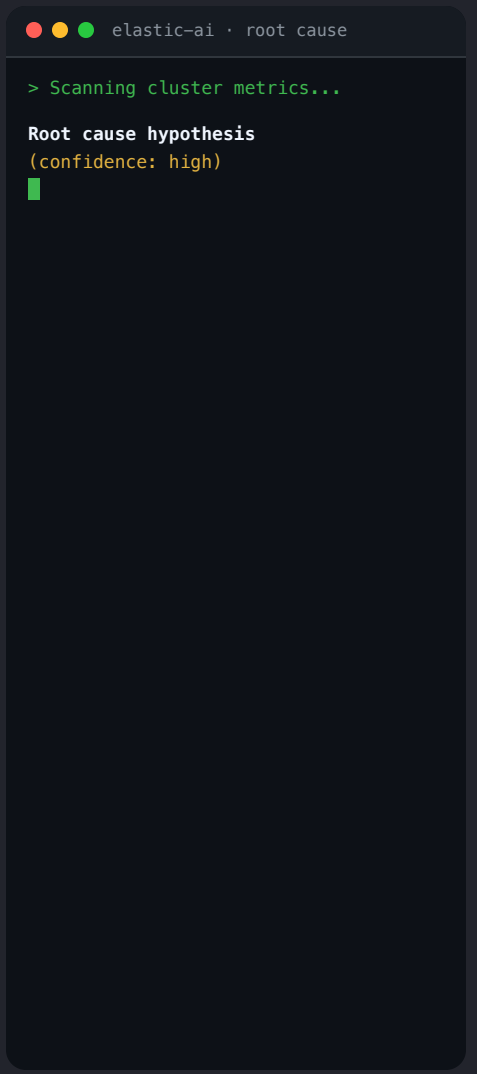
93% of companies using or evaluating Kubernetes

Customers are eager for richer monitoring and observability experiences that just work.

Enhanced OTel Experience

- Revamped Dashboards & Alerts
- OOTB SLOs & Agent Skills
- Kubernetes Workflows





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Talk to your systems in plain English

Get results in plain language, inside your existing Agents and workflows.

Native MCP server

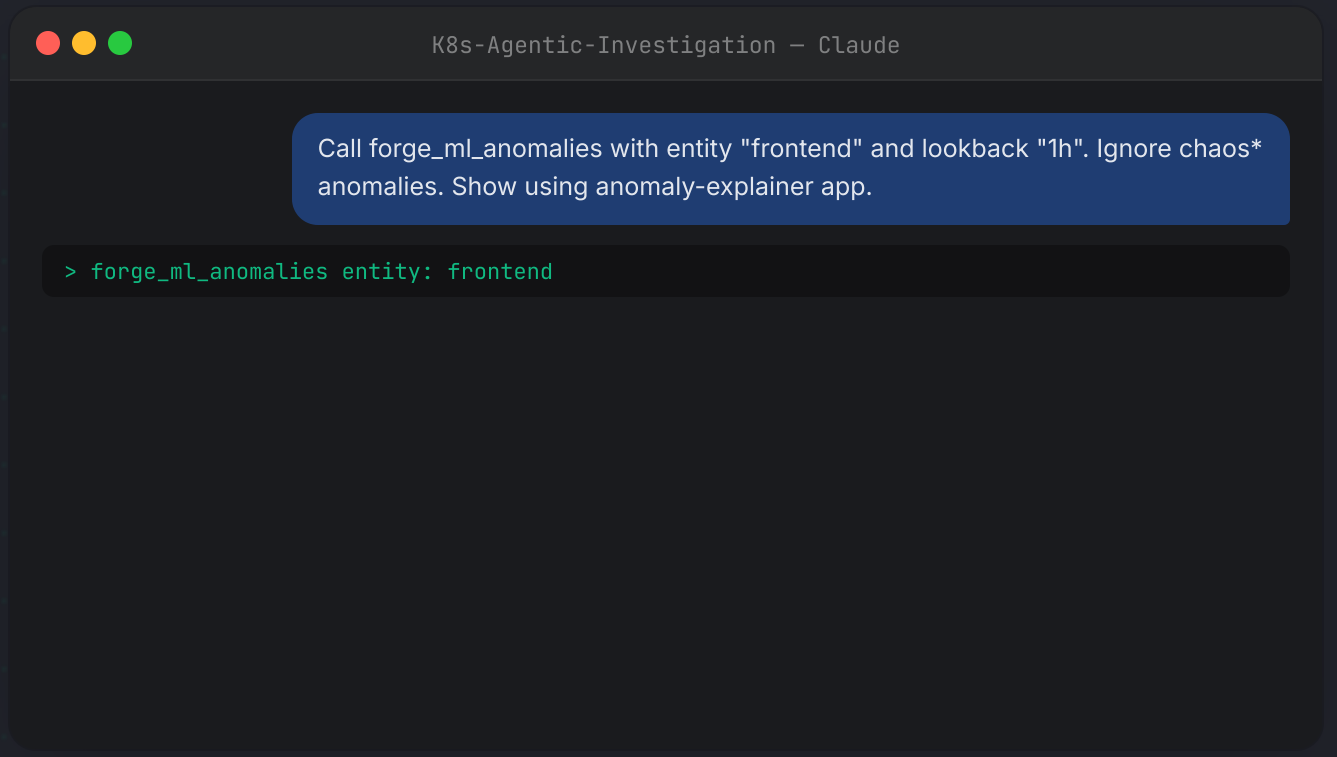
Connect any Claude-compatible client directly to your Elastic cluster.

Skills loaded automatically

Significant Events RCA, anomaly detection, and remediation — always available.

Surface-aware rendering

Results rendered as rich cards, not raw JSON — built for the interface you're using.



ELASTIC FOR CLAUDE · MCP INTEGRATION

Elastic MCP App for Kubernetes

Elastic's **MCP server** brings full observability into Claude: health dashboards, ML anomaly detection, and blast radius analysis, all through **natural language**.

- **Health Summary**
  Real-time system health across services and pods
- **Anomaly Explorer**
  ML-powered deviation detection with risk scoring
- **Blast Radius**
  Impact analysis when a service or node goes down

→ advance through the demo flow



elastic


NIGHTSHIFY

03 · PILLAR · NIGHTSHIFT

Nightshift

Autonomous AI SRE. Investigates incidents, remediates issues, and learns — around the clock, without a page.

Detect

Correlates alerts, surfaces anomalies, and ranks incidents by blast radius — before a human wakes up.

- ML anomaly detection
- Alert correlation & dedup
- Blast radius scoring

ALWAYS ON

Investigate

Runs parallel hypotheses, pulls context from every signal, and ranks findings by evidence strength.

- Parallel hypothesis engine
- Evidence-ranked findings
- Full signal context

CORE ENGINE

Remediate

Proposes and executes fixes at the trust level you define — from suggest-only to fully autonomous.

- Configurable trust levels
- Auto-fix execution
- Rollback safety

Audit

Every step cited, logged, and explainable. Full compliance trail from trigger to resolution.

- Full audit trail
- Explainable decisions
- Compliance-ready

AUTONOMOUS COVERAGE

Around the clock — no page needed

Nightshift wakes up the moment an alert fires, investigates in parallel, and acts at whatever trust level you've configured.

24/7	0	∞
continuous coverage	on-call pages	parallel hypotheses

CONFIGURABLE TRUST

You choose the autonomy level

From read-only suggestions to full auto-remediation — every action is audited, cited, and reversible.

Suggest	Confirm	Auto
read-only mode	human-in-the-loop	fully autonomous

ELASTIC OBSERVABILITY · STRATEGY

Agentic Observability

Logs *carries context* **Metrics** *carries state* **Traces** *carries both*

01

Infer Knowledge from Telemetry

No config toil: detect technologies, entities and relationships automatically from every signal stream.

02

Discover Significant Events

Stitch alerts into a **complete incident picture** before the business feels the impact: with agentic investigations.

03

Remediate with Human in the Loop

Agentic and procedural workflows that fix issues immediately, grounded in your operational knowledge.

04

Meet customers where they are

Any surface: actionable insights delivered via the interface your users already live in.

UI · Chat · Agent · API · Alerts · Mobile

STREAMS · AUTO-DETECTION

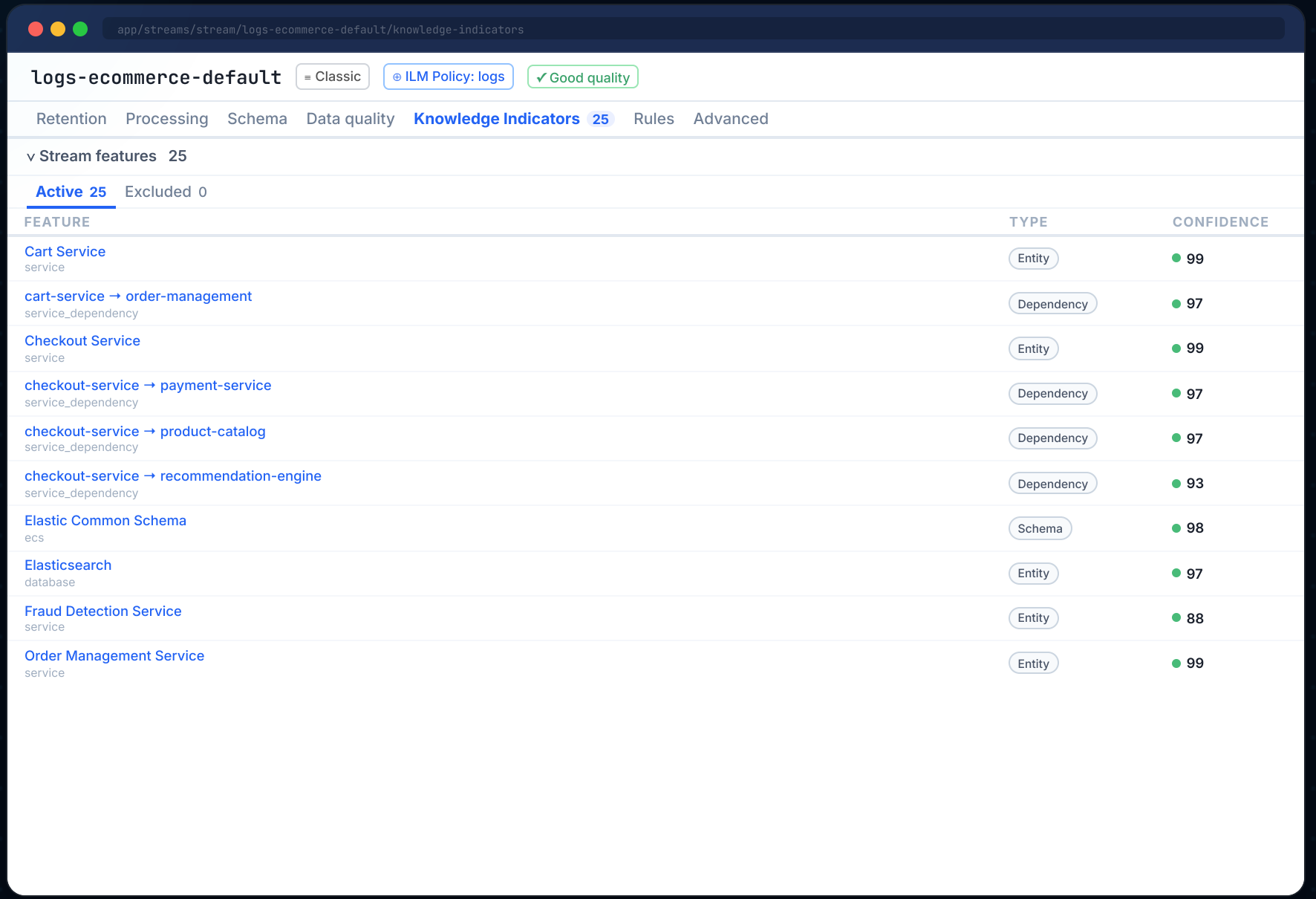
Knowledge Indicators

Entity Dependency Schema

Streams **automatically extracts** system knowledge from your telemetry: no configuration, no instrumentation required.

- **Entity**
  Services, databases & technologies detected from log patterns
- **Dependency**
  Service-to-service relationships inferred from call patterns
- **Schema**
  ECS & domain schema mappings applied automatically

Click any row to explore its dependency graph →



LIVE MODEL

The Agent's Live System Model

Built continuously from telemetry. Used to reason about incidents.

Auto-updated 3s ago

KI-DRIVEN · ALWAYS LIVE

During an incident, the agent navigates the *live model* to **understand impact** and find the **root cause.**

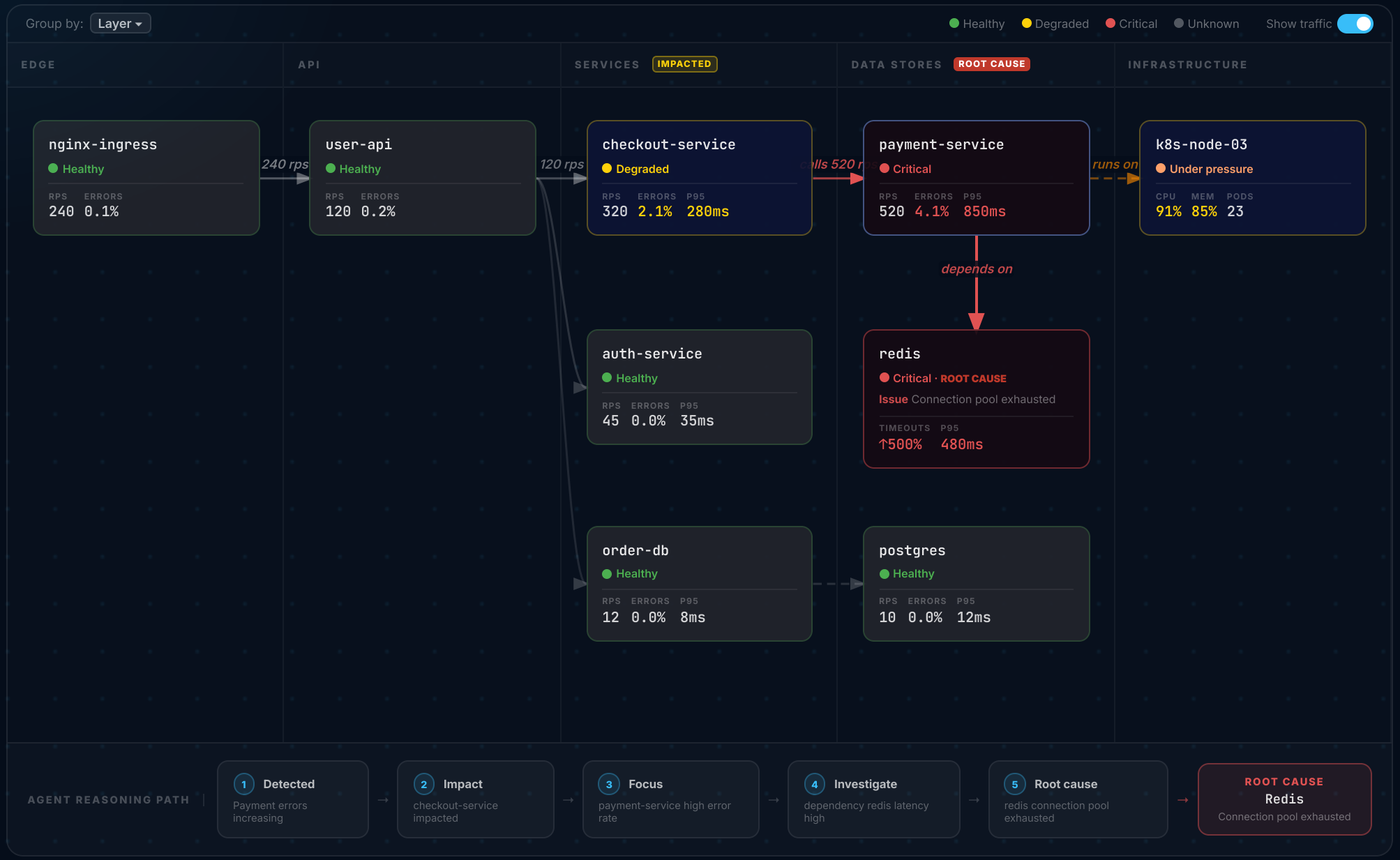
⊕
Live & continuous
Structure and health reflect real-time signals from your environment.

⟳
Agent reasoning
The agent traverses dependencies to measure impact and isolate the root cause.

◈
Knowledge indicators
Signals and patterns are extracted from telemetry and attached to relevant parts of the model.

⬡
No manual mapping
Topology is discovered automatically from telemetry, deployments and runtime behavior.

⚡
No instrumentation
Zero code changes required. The model builds itself from existing telemetry signals.



Learn how it works

ELASTIC OBSERVABILITY · DISCOVERY

Discovery: From Events to Significant Event

Events are correlated, analyzed, and transformed into a fully understood operational situation

EVENTS

Raw signals

- **OOM Exception** — Application log
- **High Latency** — Metric
- **Pod Restart** — Kubernetes event
- **Error Spike** — Service

→ + KIS

DISCOVERY

Analysis & Reasoning

INPUT SOURCES

- Events
- Knowledge Indicators

- Correlation
- Root Cause Analysis
- Blast Radius Analysis
- Event Grouping

Agent-driven investigation

→ SITUATION

SIGNIFICANT EVENT

Consolidated Situation

Memory exhaustion in service A causing cascading failures across dependent services

Root Cause

Memory exhaustion in service A

Impact / Blast Radius

Affecting service B, service C, and dependent APIs

Grouped Events

OOM · Latency · Restart · Errors

Score: High

Can trigger alert

The Discovery process correlates raw events using system knowledge to **identify meaningful patterns** and group related signals into **a single coherent operational situation.**

AGENT-DRIVEN INVESTIGATION

DEVELOPER ARC

Developer

01

Code quality and delivery — from error to merged fix.

- **Sentry**
  Error → Fix → PR
- **Perf**
  Regression → Commit
- **Deploy**
  Rollout → Safe / Stop

RELIABILITY ARC

Reliability

02

Incident operations — from alert to resolution.

- **RCA**
  Incident → Root cause
- **Triage**
  Page → Brief
- **SLO**
  Burn → Mitigation

INFRASTRUCTURE ARC

Infrastructure

03

Resource efficiency — cost, scale, and drift under control.

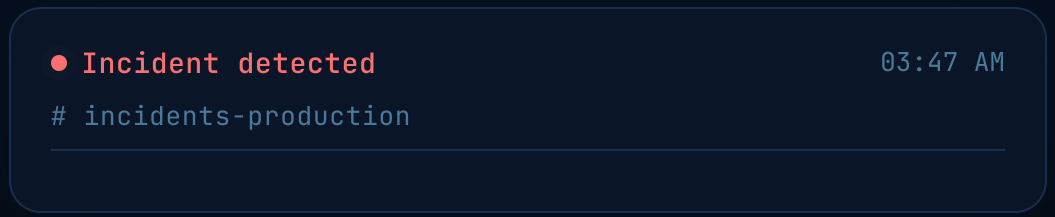
- **Cost**
  Waste → Savings
- **Capacity**
  Trend → Pre-scale
- **Drift**
  Diverge → PR

NIGHTSHIFT

Meet your AI SRE.

Always watching, always learning, always ready.

Connect your data. The rest takes care of itself.



Your system, ready

An Elastic cluster: waiting. Before a single alert, before a single connection. This is where Nightshift begins.



Published where you work. Connected to what you operate.

API-first — Kibana, IDE extensions, CLI, Slack, CI pipelines, and AI agents are all equal consumers.

DISTRIBUTION

Published where engineers actually work.

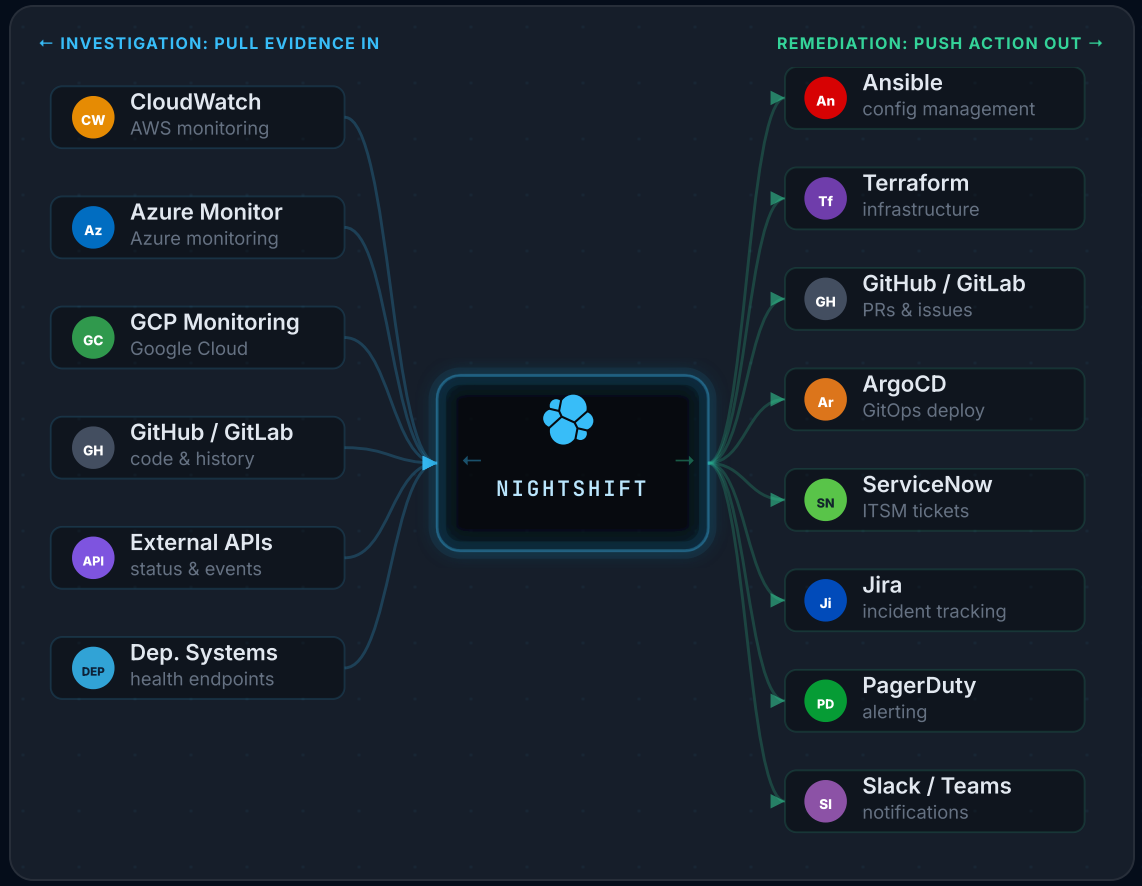
Every capability ships as an API first. Kibana is one consumer — agents, CI pipelines, terminals, and IDEs are equal ones.

IDE Extension VS Code · JetBrains · Cursor	**Kibana** dashboards · investigation · alerts	**Slack / PagerDuty** incidents · alerts · approvals
CLI / Terminal npm · pip · Homebrew		**CI / CD Gates** GitHub Actions · GitLab CI
MCP Registry Claude · Copilot · Gemini	**API / Automation** structured outputs · webhooks	**Agent Frameworks** LangChain · CrewAI · custom

Key principle: **Skills are reusable everywhere. Surfaces are just projections.**

ACTION CONNECTORS

Two types. Two directions. Never direct infra access.



Note: IngestHub handles all data-in connectors separately. These action connectors carry context and action out — never raw infra credentials.

CONTEXT LAYER · NIGHTSHIFT

The Context Layer above every O11y stack.

The Context Layer connects to the monitoring tools and data stores already in your environment — no migration required.

ELASTIC · CONTEXT LAYER

Failure Signatures · Knowledge Indicators · System Model · Pattern Library

Accumulated intelligence built from every telemetry source you own

- Failure Signatures
- Fix Outcomes
- Pattern Library
- Knowledge Indicators
- System Model
- Service Topology
- Dependencies
- Anomaly Events

RAW TELEMETRY → KI EXTRACTION · SYSTEM MODEL

ALERTS & EVENTS → SIGNAL AUGMENTATION

TELEMETRY DATASTORES

Data stays where it is. We read directly from any store.

- **Elasticsearch** — primary store · full fidelity
- **S3** — log archives · cold telemetry
- **Databricks** — lakehouse · structured events
- **BigQuery** — analytics · GCP telemetry

EXTERNAL MONITORING SIGNALS

No migration. We read alerts and anomalies directly from your existing stack.

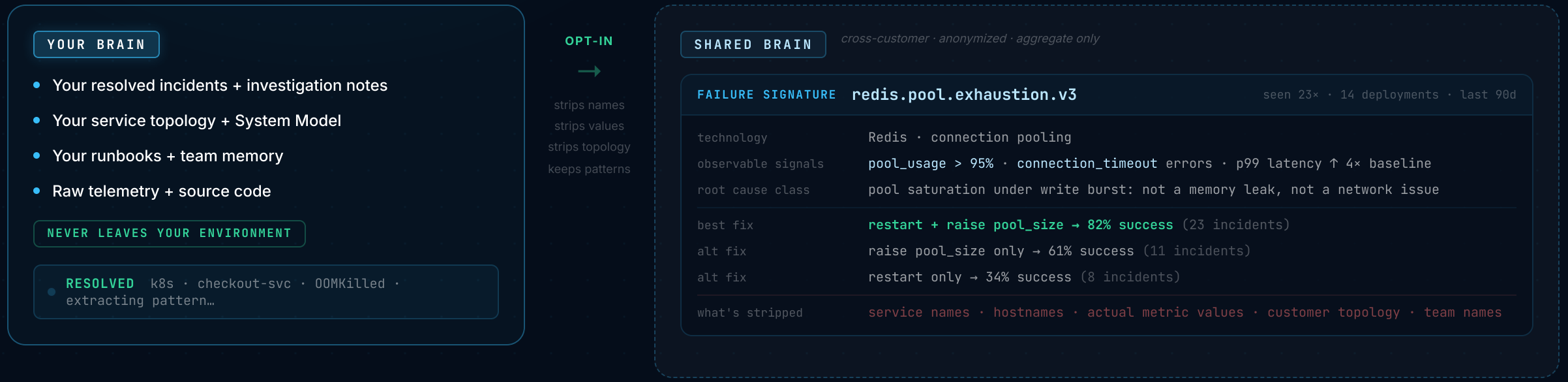
- **Datadog** — metrics, alerts, dashboards
- **Grafana** — dashboards, Loki logs
- **CloudWatch** — AWS logs & metrics
- **Dynatrace** — traces, APM alerts

Elastic Brain

What's yours. What's shared.

The memory layer behind Nightshift — learns from every incident, gets smarter every time.

Every incident your team resolves becomes a **failure signature** — an anonymized pattern the entire network can learn from. Your data never leaves. The knowledge does.



AI ECONOMICS

Token efficiency is a core differentiator.

You can't pass petabytes of raw logs to an LLM. The platform that compresses signal without losing meaning wins: on quality and cost.



Raw Telemetry

Logs, metrics, traces — logs are the primary GTM signal

Petabytes

Never passed to LLM directly. Stored at low cost, queried on demand via ES|QL.

↓ KI Extractor distills signal



Knowledge Indicators

Entities, dependencies, state, context — structured system understanding

Megabytes

The right context, structured. Agents reason over this layer for investigation.

↓ Investigation correlates



Significant Events

Correlated situations with RCA, blast radius, severity — what actually matters

Kilobytes

Precise context for action. This is what gets passed to the LLM and the human.

The hierarchy IS the token strategy. Each layer compresses meaning while discarding noise. Agents reason over Significant Events and KIs — not raw logs. This is what makes Elastic's AI economically viable at enterprise scale.

OBSERVABILITY · AGENTIC SRE

Nightshift, the Elastic AI SRE

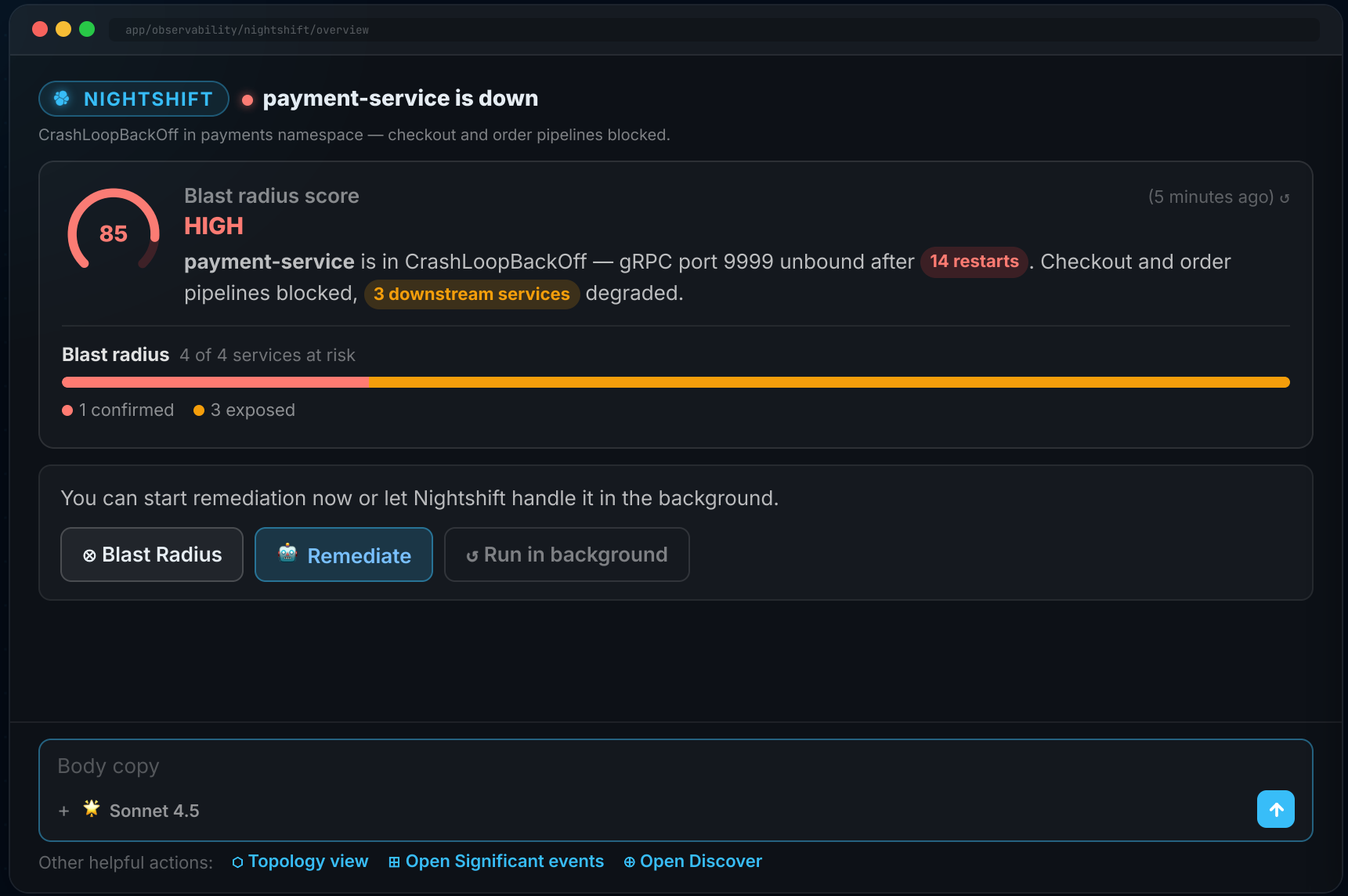
Blast Radius: HIGH | Nightshift Active | Agent Ready

MODE: Chat | Background | Agent Client

Nightshift monitors your systems while your team rests: detecting anomalies, grouping significant events, and surfacing **AI-driven remediation** the moment something goes wrong.

- **Blast Radius**
  Real-time scoring of how far an incident has spread across your system
- **AI Remediation**
  Agent-driven root cause analysis with kubectl-ready next steps
- **Background Execution**
  Run remediation autonomously: wake up to a resolved incident

Press → to play through each demo



NIGHTSHIFT

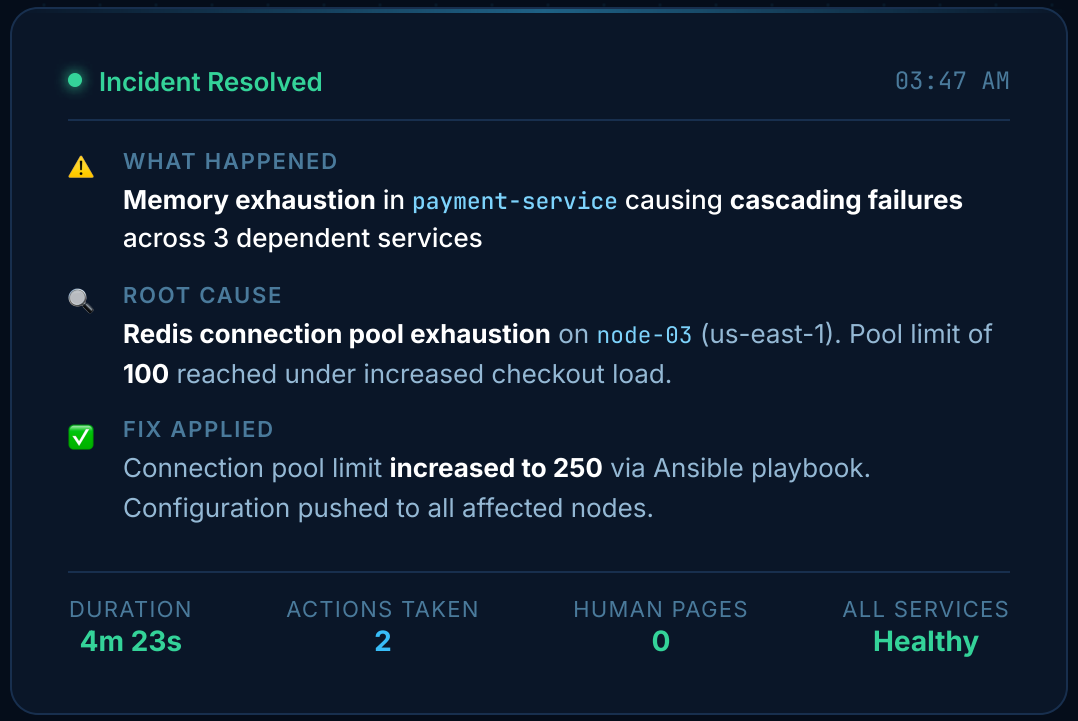
Wake up to this.

You are not buying observability.

You are buying the end of on-call.

Zero pages. Zero 3am calls.

one full explanation: ready when you are



ELASTIC OBSERVABILITY

Thank you.

Questions? Let's keep the conversation going.

elastic.co/observability



elastic

The Search
AI Company